

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.27)

Metric

$$ds^2 = \left(-\frac{e(u)^2 - 2rm(u)}{r^2} \right) du \otimes du - du \otimes dr - dr \otimes du + r^2 dx \otimes dx + r^2 dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(\frac{e(u)^2 - 2rm(u)}{2r^2} \right) du + (1) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r \right) dx + \left(\frac{1}{2} i \sqrt{2} r \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r \right) dx + \left(-\frac{1}{2} i \sqrt{2} r \right) dy$$

Contravariant components

$$l = (-1) dr$$

$$n = (-1) \, du + \left(\frac{e(u)^2 - 2 \, r m(u)}{2 \, r^2} \right) \, dr$$

$$m = \left(\frac{\sqrt{2}}{2 \, r} \right) \, dx + \left(\frac{i \, \sqrt{2}}{2 \, r} \right) \, dy$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2 \, r} \right) \, dx + \left(-\frac{i \, \sqrt{2}}{2 \, r} \right) \, dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + \frac{e(u)^2 - 2 \, r m(u)}{2 \, r^2} \, \partial_r$$

$$\delta = \frac{\sqrt{2}}{2 \, r} \, \partial_x + \frac{i \, \sqrt{2}}{2 \, r} \, \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2}}{2 \, r} \, \partial_x - \frac{i \, \sqrt{2}}{2 \, r} \, \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{e(u)^2 - 2rm(u)}{2r^3}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{e(u)^2 - rm(u)}{2r^3}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{e(u)^2}{2r^4}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{e(u) \frac{\partial}{\partial u} e(u) - r \frac{\partial}{\partial u} m(u)}{r^3}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{e(u)^2 - rm(u)}{r^4}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "